

**SECOND SEMESTER**  
**M.Tech. (AI)**

**MID SEMESTER EXAMINATION**      **March-2023**

**AI504 ARTIFICIAL NEURAL NETWORKS**

Time: 1:30 Hours

Max. Marks: 20

**Note:** Answer **ALL** questions.  
Assume suitable missing data, if any.  
CO# is course outcome related to question

1[a] Explain the functioning of the artificial neural network (ANN) with the help of a neat diagram. [2] [CO2]

[b] Obtain the output of neuron Y for the ANN shown in Fig.1, using the activation function as  $\varphi(v) = \frac{1}{1 + \exp(-v)}$ . [2] [CO2]

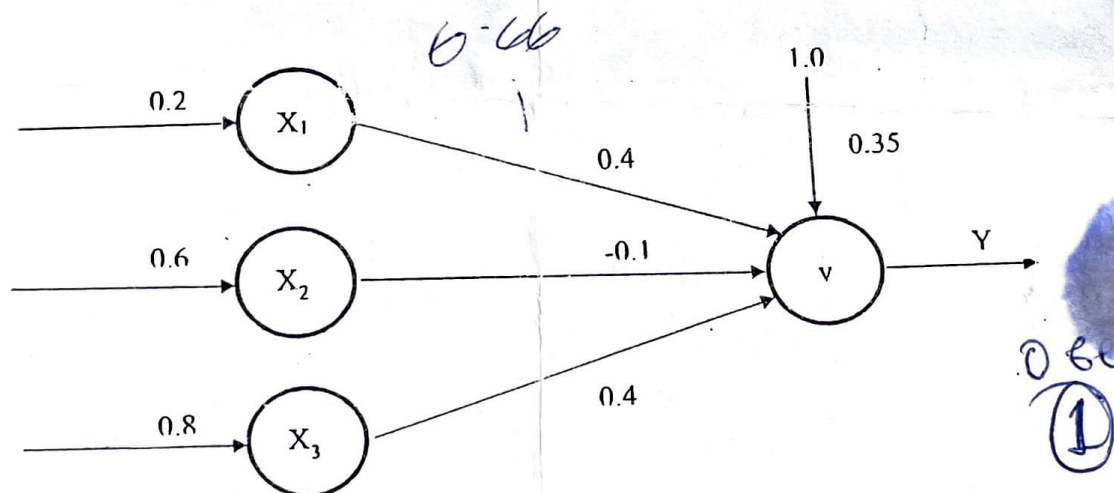


Fig. 1

2[a] Demonstrate that single perceptron can be used to realize as OR gate functions. Also, show that single layer perceptron model fails to realize XOR gate function. [2+2] [CO1]

[b] Find the value of 'w' such that the neural network shown in Fig. 2 works as XOR gate. Explain its working and show each step. Assume the activation function of neuron to be step function. [3] [CO1]

3x4

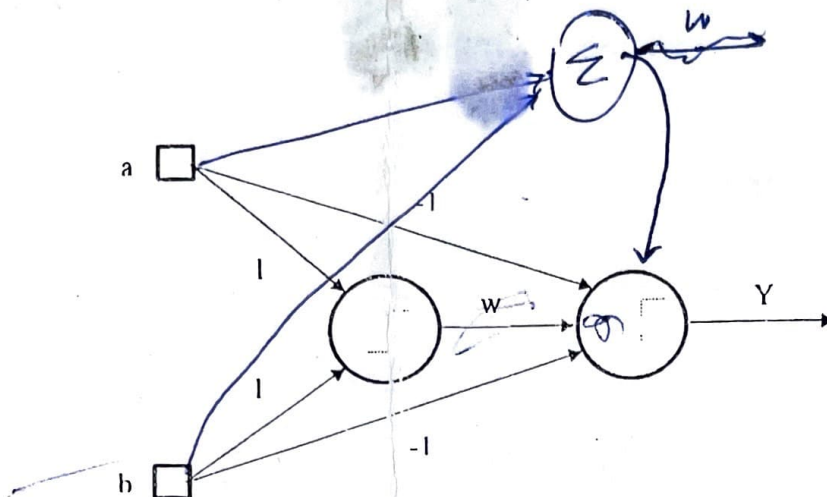


Fig. 2

- 3[a] The Fig. 3 shows a neural network. Compute the output at  $a_3^{(1)}$  and  $h_w(x)$  by using feed-forward approach. Further, obtain the expression of gradients in backpropagation at  $a_2^{(1)}$ . Let  $w_{ij}^{(l)}$  is initial weight between  $i^{th}$  neuron of layer  $l$  and  $j^{th}$  neuron of layer  $(l-1)$ . Assume the sigmoid activation function. [2+3] [CO2]

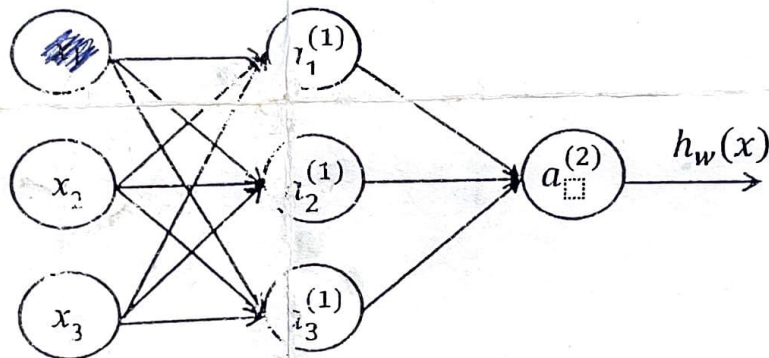
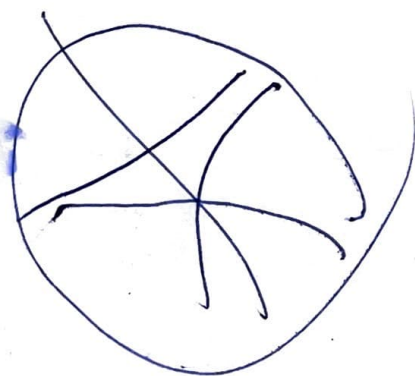


Fig. 3

- [b] Demonstrate that the cost function used in logistic regression shows the convex nature. Assume single sample. [4] [CO2]



---Best of Luck---



**END SEMESTER EXAMINATION May-2023****AI504 ARTIFICIAL NEURAL NETWORKS****Time: 3:00 Hours****Max. Marks: 40**

**Note:** Answer **ALL** questions.  
 Assume suitable missing data, if any.  
 CO# is course outcome related to question

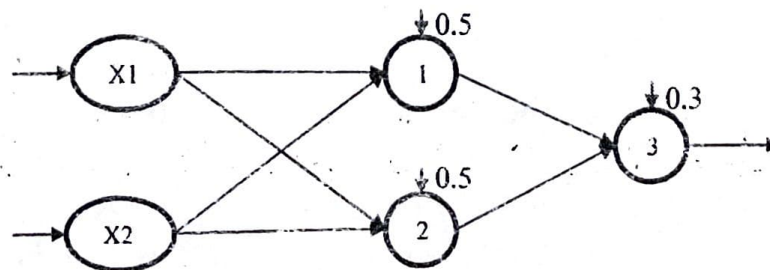
- 1[a] Consider perceptron with two input features is used to classify items as good (1) and faulty (0). Given dataset (1, 2, 1), (-1, 1, -1), (0, -1, 1), (2, 2, -1). The triplet  $(x_1, x_2, y)$  in the dataset represents features  $(x_1, x_2)$  and class  $(y)$ . The perceptron's weight vector is initialized to  $[0.1, 0.2]$ , and the bias term is initialized to 0. The learning rate is set to 0.1. Perform two iterations of perceptron training using the training data above. Show the updated weight vector and bias term after each iteration (each sample). Use the sign function as the activation function.

**[4] [CO1, CO2]**

- [b] The sigmoid function is used as activation function in logistic regression. It makes mean squared error based cost function unsuitable for gradient decent, explain why? Give an alternate cost function, prove that it is suitable for gradient decent algorithm.

**[4][CO2]****2. Attempt any two**

- [a] Consider the neural network (shown in Fig.1) to predict the binary output  $y$  based on two input features  $x_1$  and  $x_2$  using the given dataset  $(x_1, x_2, y)$ : (1, 2, 0) (1, 1, 0) (4, 3, 1) (5, 4, 1).

**Fig. 1**



Assume ReLU activation for the hidden layer, and sigmoid activation for the output layer. Initialize the weights of the hidden layer to be  $W^{[1]} = [[0.1, -0.2], [0.3, -0.4]]$  and the bias to be  $b_1 = [0.5, 0.5]$ . Initialize the weights of the output layer to be  $W^{[2]} = [[0.2], [-0.1]]$  and the bias to be  $b_2 = [0.3]$ . Find predicted output by the neural network. [4] [CO2]

- [b] Assume that you are working on a project to develop a spam filter for emails using a perceptron. The perceptron takes as input a bag-of-words representation of the email's text, where each element of the input vector corresponds to the count of a particular word in the email. We are given a dataset of 1000 emails, half of which are spam and half of which are not spam, and we randomly split the dataset into a training set of 800 emails and a validation set of 200 emails. We initialize the perceptron's weight vector and bias term to all zeros, and set the learning rate to 0.1.

(i) Suppose the perceptron achieves an accuracy of 90% on the validation set. Does this mean that the perceptron is a good spam filter? Explain your reasoning.

(ii) Now, suppose the given dataset 90% spam email and 10% as not spam. Does 98% accuracy on the validation set indicate a good spam filter? Explain your reasoning. [2+2][CO2, CO3]

- [c] In above spam-classifier design, suppose we decide to experiment with different learning rates to see if we can improve the perceptron's performance. We try learning rates of 0.01, 0.1, and 1.0. For each learning rate, we train the perceptron using the training set and evaluate its performance on the validation set. We find the following results:

| Learning rate              | 0.01 | 0.1 | 1.0 |
|----------------------------|------|-----|-----|
| Accuracy on validation set | 92%  | 95% | 89% |

Based on these results, which learning rate should we use for the perceptron and why? [4][CO2, CO3]

### 3. Attempt any two

- [a] Consider training of the neural network shown in Fig.1. Let the cost function is binary cross-entropy. Perform one iteration of backpropagation using the first data point to update the weights and biases. Use a learning rate of 0.1. Calculate the updated weights and biases of the output layer after one iteration of backpropagation.

[4] [CO2]



[b] Derive the expression for weight updation (backpropagation) of the weight  $w_{11}$  i.e. the weight between input  $x_1$  and neuron 1 of the hidden layer. Show each step clearly with computation graph. [4] [CO2]

[c] Consider a neural network with three output nodes, and the output values from the last layer before the softmax activation are [2.5, 1.0, 0.7]. Calculate the softmax values for each output node. What is the index of the node with the highest softmax value? What is the interpretation of the softmax values? [4] [CO2]

4. Attempt any two

[a] Suppose you are working on a classification problem with a dataset consisting of 10,000 samples. You have trained three models with the following accuracy values:

Model A: training accuracy = 85%, validation accuracy = 80%

Model B: training accuracy = 90%, validation accuracy = 85%

Model C: training accuracy = 95%, validation accuracy = 80%

(i) Which model(s) suffer from high bias and which model(s) suffer from high variance?

(ii) What are the possible reasons for the observed bias and variance in each model? [4] [CO1, CO2]

[b] Explain the trade-off between bias and variance in machine learning and how it can be managed. How would you go about choosing the best model for above (in part [a]) particular classification problem?

Now, suppose you have access to more data and can double the size of the dataset to 20,000 samples. How do you think this would affect the bias and variance of the models? How would you use this additional data to improve the performance of the models? [2+2] [CO1, CO2]

[c] An artificial neural network is designed for binary classification of a disease. The performance is evaluated on 2000 patients. The confusion matrix for the classification results is as follows:

|                  | Predicted Positives | Predicted Negatives |
|------------------|---------------------|---------------------|
| Actual Positives | 300                 | 100                 |
| Actual Negatives | 400                 | 1200                |

Calculate the accuracy, recall, precision, and F1 score of the model for the positive class. Discuss the importance of each measure, if the disease is highly contagious. [4] [CO2, CO3]



5. A convolutional neural network (CNN) has the following architecture:  
*Input layer*:  $32 \times 32 \times 3$  image; *Convolutional layer*: with 64 filters, each with size  $5 \times 5$ , using stride 1 and no padding; *Max pooling layer* with size  $2 \times 2$ ; *Convolutional layer* with 128 filters, each with size  $3 \times 3$ , using stride 2 and no padding; *Max pooling layer* with size  $2 \times 2$ ; *Convolutional layer* with 256 filters, each with size  $3 \times 3$ , using stride 1 and full padding; *Max pooling layer* with size  $2 \times 2$ ; *Fully connected layer* with 512 neurons; *Output layer* with 10 neurons.

[a] Find the size feature maps before fully connected layer. Show the calculation for each step. [4] [CO2, CO3]

[b] Compute the number of learnable parameters in each layer of the above CNN architecture. [4] [CO2, CO3]

---Best of Luck---